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Machines vs. humans: The evolving role of artificial intelligence in livestreaming e-commerce

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ABSTRACT

As the capability of artificial intelligence (AI) improves, online retailers are exploring AI-based agents to communicate with viewers in live streaming, which is referred to as AI stremer. However, it is unclear where, what, and when the implementation of AI stremer is more effective than human beings in live-streaming e-commerce. To explore the dynamic interrelationships and temporal evolution between AI and human streamers and viewer engagement, this study examined the evolving role of AI streamers in live-streaming e-commerce. We utilised the linear mixed model (LMM) and the time-varying effect model (TVEM) to examine whether AI and human streamers differ in both monetary and non-monetary engagement activities. Additionally, we investigated how these differences change over time and whether such changes are consistent across different consumption contexts. The dataset consists of 924,036 products from 21,190 live streaming shows in 123 live broadcasting rooms over a period of four months was used in this study. The results suggest that AI streamers can substitute for humans in monetary activities in the context of utilitarian consumption but not in hedonic consumption. However, the substitute effect of AI may gradually diminish over time. In addition, in a hedonic context, AI exhibits an increasing effect on viewer engagement over time.

"We are in a rare position to create change. To reinvent what it means to be a marketer. This is your chance to be a pioneer. Don't wait for the marketing world to get smarter around you. Take the initiative now to understand, pilot and scale AI."
—Paul Roetzer, founder of the Marketing Artificial Intelligence Institute

1. Introduction

Livestreaming — an emerging social media platform that facilitates real-time interaction among participants — has experienced a rapid surge in popularity worldwide (Hilvert-Bruce et al., 2018). For instance, the livestreaming e-commerce market in China reached approximately 1,237.9 billion yuan in 2020, showing an annual growth rate of 197.0 %. The market size of livestreaming commerce is projected to exceed 4.9 trillion yuan by 2023.[1] In the United States (US), livestreaming e-commerce is expected to generate online sales worth $11 billion in 2021, while revenue from live shopping is projected to triple, reaching $35 billion by 2024.[2] The widely acknowledged transformative potential of livestreaming is reshaping the future landscape of online shopping (Bloomberg, 2020).

Livestreaming distinguishes itself from traditional online marketing approaches by facilitating real-time interaction between streamers and viewers (Meng et al., 2021). Specifically, during livestreaming sessions, streamers deliver comprehensive introductions to the products they are selling, emphasising their features, benefits, and exclusive pricing offers. At the same time, viewers can engage with the streamers in various ways during the livestreaming session. Some engagement activities are monetary, such as purchasing the product and tipping the streamers with tokens or gifts that viewers buy with real money on the platform (Guan et al., 2022). Additionally, there are non-monetary engagement

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[1] https://www.statista.com/statistics/1127635/china-market-size-of-live-commerce/.
[2] https://www.statista.com/statistics/1276120/livestream-e-commerce-sales-united-states/.

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activities (Lin et al., 2021), such as chatting with *danmaku*,[3] sending 'likes,' and following streamers. Streamers can respond to viewers immediately after viewing the *danmaku*. These distinctive features in livestreaming shape viewer engagement and are widely recognised by firms, as well as rigorously scrutinised by academics (Hu & Chaudhry, 2020).

As a result, companies have started cultivating streamers. For instance, in 2020, the number of employed livestreamers in China reached approximately 12.34 million, showcasing a remarkable growth rate of over 350 % compared to the previous year.[4] Despite this rapid expansion, there is still a substantial demand for livestreamers within the industry. Projections indicate that there will be employment gaps of 8 million in the field of livestreaming in China in 2021, and the number is expected to increase to 19.415,000 by 2025. Additionally, despite significant variations in performance, the costs associated with human streamers have increased (Wang et al., 2021).

In response to the widening gap between demand and supply in livestreaming, firms have increasingly turned to artificial intelligence (AI) technology as an alternative solution. As an example, the Taobao platform has recognised AI streamers and virtual scenes as two highly promising avenues for the development of livestreaming e-commerce in 2022.[5] Moreover, in Baidu World 2022 conference, Baidu AI Cloud introduced a digital livestreaming platform that leverages AI. Compared to its human counterparts, AI can reduce the cost of livestreaming by over 30 %.

AI can be seen as the affordance of human intelligence to machines (Ma and Sun 2020). AI streamers, as highlighted by Wang et al. (2021), provide viewers with detailed product information and exclusive coupons. They are also capable of addressing viewers' inquiries during livestreaming. AI streamers offer advantages such as cost-effectiveness, remarkable flexibility, and job permanence.[6] Traditionally, emotions and the mind were perceived as intricate human response systems. However, advancements in cognitive technology and ongoing functionality have shifted this perspective, suggesting that AI endowed with humanoid features may possess emotional capabilities (Vuong et al., 2023a). This notion has led to a belief that AI has the ability to exhibit traits akin to human cognition (Vuong et al., 2023b).

However, AI streamers are not without limitations. For instance, they may encounter challenges in addressing issues beyond predefined topics and lack the ability to provide personalised interactions with viewers—a capability inherent to human streamers due to their unique social skills (Wang et al., 2022). Moreover, the rise of ethical concerns in AI, including issues like consumer alienation, has sparked renewed interest among consumers regarding unmediated social connections, such as authentic consumption experiences (Puntoni et al., 2021). In conclusion, given the respective strengths and weaknesses of human and AI streamers, it remains uncertain whether and when the implementation of AI or human streamers will yield the most favourable outcomes in livestreaming e-commerce.

Marketers seeking answers to these questions will discover limited guidance in the academic literature. Despite recent studies investigating the relationship between humans and AI in the fields of management and information systems (Garvey et al., 2022; Lyytinen et al., 2020), particularly within the context of livestreaming e-commerce (Wang et al., 2022; Xu & Ruan, 2023), a comprehensive consensus regarding the distinctions, boundaries, and dynamic fluctuations between AI and human streamers is currently lacking. Indeed, in its literature survey on AI marketing, a paper by McKinsey Global Institute emphasised that "the impact of AI might not be linear but could build up at an accelerating pace over time."[7] The relationship between AI and humans can function as a substitute or an augmentation, contingent upon temporal and spatial factors (Manis & Madhavaram, 2023; Raisch & Krakowski 2021).

The issues discussed above motivated the initiation of this investigation. To address concerns regarding the suitability and applicability of AI streamers in livestreaming e-commerce, this study aimed to assess the impact of both AI and human streamers in various consumption contexts over time. Specifically, this research sought to answer the following questions:

1. To what extent and in which fields do AI streamers outperform human beings in livestreaming?
2. How do the effects of AI and human streamers on viewer engagement vary over time in livestreaming?

This study aimed to address these questions by incorporating an innovative temporal scale perspective. In livestreaming e-commerce, viewers express their interactions through monetary and non-monetary engagement activities. Additionally, viewers' preference for human beings or AI streamers depends to the consumption context (utilitarian or hedonic). The consumption context is likely to affect viewers' engagement in livestreaming ecommerce. Therefore, we utilised a linear mixed model (LMM) and a time-varying effect model (TVEM) to examine whether AI and human streamers differ in monetary and non-monetary engagement activities, how this difference may change over time, and whether this change is consistent across consumption contexts.

Utilising data collected from retailers implementing both AI and human streamers in livestreaming e-commerce on the Taobao platform, we observed that AI adoption led to interrelated dynamics of substitution and augmentation alongside humans in various consumption contexts. Firstly, we found evidence that AI streamers can substitute for their human counterparts in monetary engagement activities in livestreaming but do not perform as well in non-monetary engagement activities. Secondly, the substitution effect in the utilitarian context was short-lived. As more retailers incorporate AI streamers into livestreaming, the competitive advantage of being the first to introduce AI streamers gradually diminishes. Thirdly, over time, augmentation becomes apparent in the hedonic context, and AI streamers excel in this particular setting.

This research contributes to the literature in three ways. Firstly, it expands the current human–machine literature by addressing the previously unexplored question of when AI streamers outperform their human counterparts in livestreaming e-commerce. Specifically, we identify the diverse effects of human beings and AI streamers in both monetary and non-monetary engagement activities. Secondly, by incorporating a time scale, this study explores the short-term and long-term effects of AI in livestreaming e-commerce. Thirdly, by examining the moderating effect of consumption context, this work reveals the important role of hedonic/utilitarian attribution in influencing the effectiveness of AI in livestreaming. Overall this study also offers new insights to enhance our understanding AI streamers in practice.

[3] *Danmaku* is a real-time commentary system that displays viewers' comments overlaid on top of the video. The comments move dynamically from one side to the other as the video plays, providing a real-time and interactive communication experience (Zhang and Cassany, 2020).

[4] https://www.statista.com/statistics/1295791/china-number-of-live-commerce-streamers/.

[5] Quarterly meeting of Taobao live MCN institutions held on April 14, 2022.

[6] According to 51job.com's recruitment information, the average monthly salary of a human is about 10,000 to 20,000 yuan. The price of the virtual live broadcast room launched by BlueFocus (https://www.bluefocusgroup.com/) and Alibaba Damo Academy is 99,000 yuan/year (BluCursor.com).

[7] Bughin et al. (2018) argued that the adoption and absorption of AI is likely to follow an S-curve pattern—starting slowly due to the substantial costs and investment associated with learning and deploying these technologies, then accelerating driven by the cumulative effect of competition, improvements in complementary capabilities, and process innovations.

2. Literature review and theoretical foundation

2.1. Livestreaming e-commerce

Livestreaming e-commerce is a subset of e-commerce embedded with real-time social interaction via livestreams (Wongkitrungrueng et al., 2020). In the early days of livestreaming commerce, platforms often promoted events by inviting social media influencers or stars (Internet celebrities), or directly cooperating with brands to promote products (Cai & Wohn 2019). Later, due to the high costs associated with influencers' or stars' livestreaming, retailers and brands began cultivating their own livestreaming teams. Dedicated brand streamers, as identified by Zhang et al. (2022), emerged, particularly in the official flagship stores of certain brands, gradually becoming the main force in livestreaming e-commerce.[8] Livestreaming e-commerce typically utilises three types of channel: e-commerce platforms integrating livestreaming features (e.g., Taobao), livestreaming platforms incorporating commercial activities (e.g., Liveme), and social networking sites that incorporate livestreaming features (e.g., Facebook Live) (Wongkitrungrueng et al., 2020). In this study, we focused on the first type of channel.

In comparison to traditional e-commerce, livestreaming commerce offers several distinct advantages. Firstly, direct and synchronous communication is a key feature of livestreaming. The streamers and viewers can interact in real-time, despite being spatially separated. This introduces social attributes to e-commerce, placing a strong emphasis on fostering and maintaining consumer relationships (Zhang et al., 2022). Secondly, unlike the text-to-text communication in traditional e-commerce, livestreaming e-commerce allows the streamer and viewers to engage in virtual face-to-text communication (Wongkitrungrueng et al., 2020). The presence of a face in livestreaming has been shown to positively impact sales (Bharadwaj et al., 2022). Moreover, livestreamers have the capability to respond to personalized questions from viewers (Zhang et al., 2022). Additionally, other viewers can also receive and engage with the replies, fostering a communal and interactive environment.

Despite the rapid development of livestreaming e-commerce in recent years, research in this area is still in its infancy, and this new phenomenon has received limited attention (Park and Lin, 2020). Most extant studies have focused on the following three perspectives (see Fig. 1):

(1) From the perspective of place, several scholars have examined the influence of presence (e.g., social presence, telepresence, and co-presence) and immersion on consumers' purchase intentions during livestreaming (Diwanji et al., 2020; Lim et al., 2015; Ming et al., 2021). With the advancement of information and communication technology, the metaverse has emerged. Barta et al. (2023) conducted a comparative analysis to assess the impact of two communication channels—live shows on Instagram and the metaverse—on followers' experiences and behaviours during livestream shopping. (2) From the perspective of products, some researchers have concentrated on examining how factors such as fit, perceived product quality uncertainty, and value influence consumer behaviour in livestreaming. For example, Park and Lin (2020) identified a positive relationship between self–product fit and purchase intention. Additionally, they observed that the product fit of Wanghong (Internet celebrities) positively affected both their trustworthiness and attractivenessFurthermore, they found that live content–product fit influenced both utilitarian and hedonic attitudes towards the live content. Lu and Chen (2021) found that product fit uncertainty and product quality uncertainty were negatively related to consumer purchase intentions in livestreaming e-commerce, while Wongkitrungrueng and Assarut (2020) revealed that utilitarian value positively affected product trust. (3) From the perspective of people, there are two aspects: consumers and livestreamers. For the former, some research has investigated the antecedents of consumer engagement, impulse buying behaviour, consumption intention, and pay-what-you-want in livestreaming e-commerce (Hou et al., 2020; Hu and Chaudhry, 2020; Lo et al., 2022; Lu et al., 2021; Ming et al., 2021). For the latter, some scholars have examined the impact of human streamers' emotions (e.g., happiness, sadness, surprise, anger, fear, disgust) (Bharadwaj et al., 2022; Lin et al., 2021), personality traits (Zhao et al., 2021), genre (e.g., fixed-brand streamers, multi-brand streamers) (Zhang et al., 2022), and characteristics (Guo et al., 2022).

Recently, scholars have begun to explore the impact of AI streamers on viewers' behaviours. For example, Zhang et al. (2023) found that the congruence between self and AI streamers exerted a significant influence on impulsive buying tendencies, but it did not directly impact intentions for parasocial interaction. These intentions were instead shaped by the perceived coolness of AI streamers. Gao et al. (2023) explored the influence of virtual streamer characteristics, such as likeability, animacy, and responsiveness, on purchase intention through the mediating factor of presence. However, only a few scholars have initiated explorations of the comparative effects of AI and human streamers in livestreaming e-commerce. For example, Wang et al. (2022) examined whether AI streamers are as effective as human streamers in livestreaming e-commerce. However, those studies did not provide empirical analysis. Zhang et al. (2023) investigated the mechanism of a virtual anchor driven by AI–human collaboration (assist vs. supervise) on consumer engagement. Xu and Ruan (2023), using an online survey, identified that AI broadcasters were better suited for consumers experiencing high levels of social overload. Gao et al. (2024) examined how the type of streamer influences consumer purchase intention.

While prior studies have documented the impacts of both AI and human streamers on consumer behaviour during livestreaming, they have not paid enough attention to how these effects change over time, which is important for the long-term development of livestream commerce. The livestreaming industry is experiencing continued growth, driven by advancements in AI, resulting in a rising presence of AI streamers on livestreaming platforms. In the context of livestreaming e-commerce, we propose that both AI and human streamers dynamically play pivotal roles in fostering consumer engagement.

2.2. Utilitarian and hedonic characteristics of consumption

Consumption activities involve utilitarian and hedonic elements. Hedonic consumption is characterized by its multisensory nature and its effective drive, providing experiential pleasure, fun, or excitement. In contrast, utilitarian consumption is predominantly instrumental and driven by cognitive motivations, focusing on functional product aspects such as performance parameters, materials, and price (Khan et al., 2005).

Previous research on hedonic and utilitarian consumption has explored the effects of recommendation systems within e-commerce (Chiu et al., 2014; Liu et al., 2020). For instance, hedonic purchases often involve online product recommendation agents with avatar interfaces, social media, and product page views on the retailer's website, incorporating social cues. In contrast, utilitarian purchases tend to rely on online product recommendation agents with explanation facilities, search engines, third-party reviews, deal sites, and product page views on competing retailers' websites, especially closer to the time of purchase (Li et al., 2020). The growing interest in recommendation systems has led scholars to focus on AI recommendations in both functional and hedonic consumption contexts. The results find that pursuing utilitarian goals makes people assign greater weight to AI recommendations, and hedonic choices are more associated with human recommendations (Longoni & Cian 2022).

Research indicates that utilitarian (hedonic) goals lead people to

[8] The turnover of fixed brand streamers accounted for 32.1% of the total livestreaming e-commerce in 2020, and this proportion is expected to rise to approximately 56.1% by 2026.

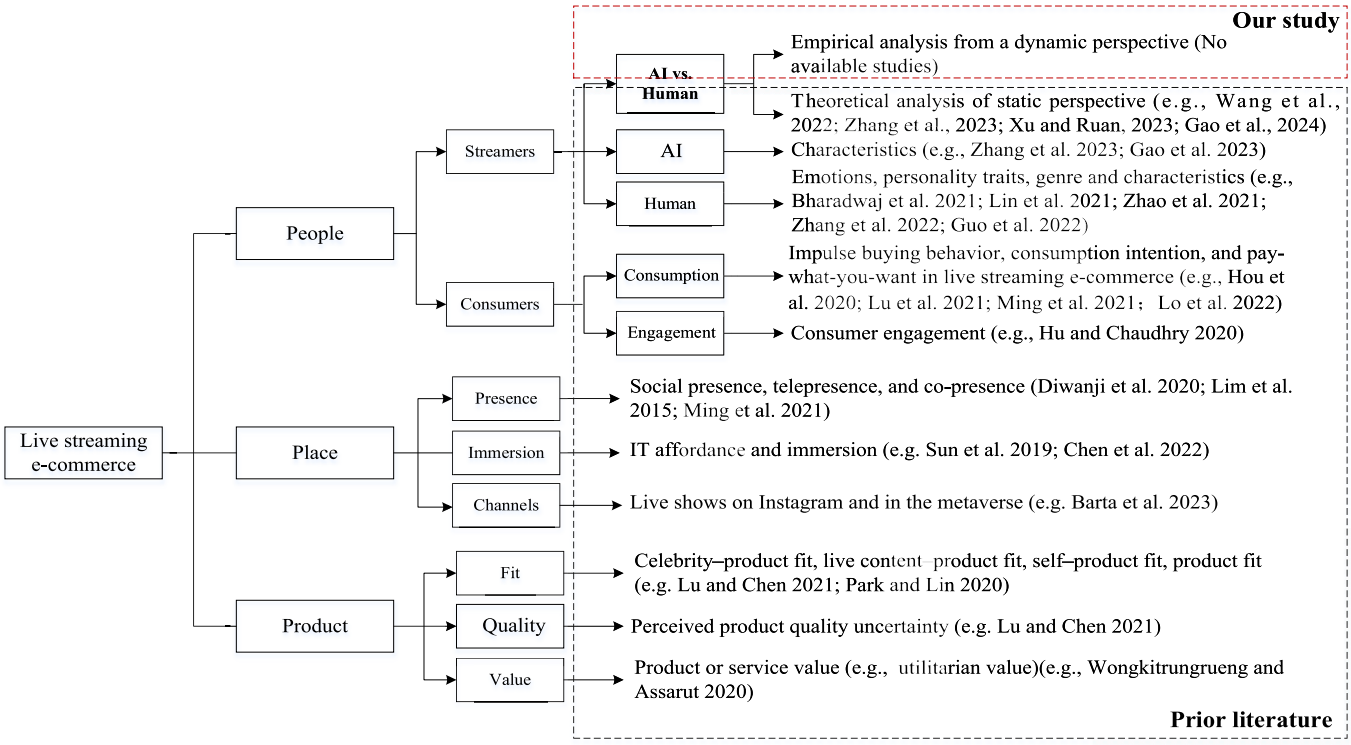


Fig. 1. Positioning our study in the livestreaming literature.

place greater emphasis on AI recommendations, whereas hedonic choices are more commonly associated with human recommendations (Longoni & Cian, 2022). Additionally, personalized recommendation strategies generated by AI-driven interactive agents have been found to be more effective for consumers seeking comfort (Kim et al., 2022). However, research on hedonic and utilitarian consumption has yet to address whether the trade-offs between these types of consumption influence the effectiveness of 'salespersons' (AI vs. human) over time. This question is gaining prominence due to its importance for livestreaming, especially in leveraging the potential of AI.

2.3. Relevant literature on AI applications

This study is related to several streams of research. First, it is related to research scrutinizing the relationship between AI and human beings in management. At the heart of this field lies a central question: Does AI substitute or complement the shortcomings of human beings? The existing literature in this stream can be categorised into three lines of research. The first line of research focuses on understanding the complementarity of machines to humans and the conditions under which complementarity occurs. A common finding from current research is that AI has the potential to enhance human abilities in certain tasks. For example, Luo et al. (2019) conducted a field experiment which demonstrated that undisclosed chatbots were equally effective as skilled workers, and four times more effective than less experienced workers. However, disclosing the chatbot's identity before the machine–customer conversation resulted in a reduction in the purchase rate by more than 79.7 %. When introducing robots into a specific service environment, managers grapple with intricate decisions across various categories, encompassing robot design, customer characteristics, and service encounter characteristics (Belanche et al., 2020a). In a series of randomised field experiments, Luo et al. (2021) discovered that the incremental returns of AI over human managers exhibited an inverted U-shaped pattern across agents. While the middle-ranked agents demonstrated the most significant performance improvement, both the lowest-ranked and top-ranked agents exhibited limited incremental gains. Garvey et al. (2022) found that due to the absence of selfish intentions, consumers respond better when dealing with an AI agent in cases where a product or service offer is worse than expected. On the contrary, for an offer that is better than expected, consumers respond more positively to a human agent. A study conducted by You and Robert (2023) demonstrated the trust and preference of co-workers towards collaborating with robots. The findings revealed that gender dissimilarity exerted a robust negative impact on the rapid establishment of trust in robot co-workers, surpassing the influence of work style and personality traits.

The second line focuses on substitution. Belanche et al. (2020b) demonstrated that frontline robots exhibit a higher degree of corporate representation than human employees. Moreover, AI has been shown to be capable of substituting human cognitive capabilities (Balasubramanian et al., 2022) and emotional support, such as providing empathetic and reassuring expressions (Gelbrich et al., 2021), owing to the benefits of machine learning. These benefits include enormous computational superiority (Choudhury et al., 2020) and better prediction than humans (Smith, 2019).

The third line of research has emerged more recently, adopting a more comprehensive perspective to explore the relationship between AI and humans. This includes Raisch and Krakowski's (2021) use of a paradox theory perspective and Krakowski et al. (2023) taking a question-based approach. Raisch and Krakowski (2021) found that the dual nature of AI applications—augmentation and automation—is interdependent across time and space. Krakowski et al. (2023) used chess as a controlled setting to examine the complex causal relationship between AI and performance. They found that the adoption of AI triggers interrelated dynamics of substitution and complementation.

Most of these scholars have conducted qualitative or experimental research from a static perspective, without examining the relationship between AI and human over a period of time. Although a few studies proposed to analyse the relationship between AI and human beings from the perspective of a time scale factor, empirical verification has not been conducted. Therefore, considering the existing literature, our model will incorporate the time factor to quantify the effects of AI applications, which have been lacking in previous research.

The second stream is related to the growing research on machine learning in the field of information system studies. Some studies conceptualise machine learning as a specialised kind of computational process (Domingos, 2012) and assume that machines either do not change or change slowly, and that their learning processes differ from those of humans (Venkatesh et al., 2003). However, this assumption overlooks the capabilities of machine learning. Lyytinen et al. (2020) noted that machines have acquired the ability to learn in a manner similar to human trial-and-error learning. Machine learning systems are constantly evolving. For instance, the advancement of technology expands the capabilities of AI beyond content creation to real-time adaptation and control (Du et al., 2024), thereby demonstrating the evolution of learning machines. AI applications will undergo an evolutionary process, manifesting functionalities beyond the explicit articulation of their designers. This transformative progression has the potential to reshape interrelationships among actors, necessitating the investigation of AI over time (Anthony et al., 2023). The exploration of temporal dynamics is crucial for comprehending the role of AI (Vaid et al., 2023). Thus, this study adopts an evolving perspective to study the role of AI in livestreaming e-commerce, echoing the call of existing literature for machine learning research in settings where metahuman systems evolve (Lyytinen et al., 2020) and exploring how humans respond to artificial intelligence at different stages (Flavian & Casaló, 2021).

This study is also linked to research that explores the roles of consumption context in online shopping settings. Previous research has established that people may prefer AI recommendations in the utilitarian realm (Longoni & Cian 2022). For example, Ruan and Mezei (2022) found that AI chatbots perform better when the product attribute is functional. Previous research also examined the impact of AI levels on decision makers' attitudinal responses and behaviours. For example, Gursoy et al. (2019) revealed that AI assistants' advantages affect users' perception of utilitarian/hedonic value, which further influences their willingness to accept AI assistants. Huang and Rust (2021a) noted that hedonic service could benefit from feeling AI, which aids in customer understanding, positioning, and relationship-building. On the other hand, utilitarian services are suited for thinking AI, which is for market analysis targeting (marketing strategy) and personalisation (Hermann, 2022). In addition, from the perspective of information processing theory, biological information processing systems update themselves over time by exchanging data with the environment (Vuong et al., 2023a). Livestreaming provides conditions for real-time information interaction during online shopping, enabling the updating of personal information processing systems. By assessing the balance of cost and benefit of the newly acquired information, people's mindsets are formed and influence their attitudes and behaviour. Using longitudinal data, this research contributes to the literature by documenting the consumption context in livestreaming. It also documents the effects of the consumption context in AI livestreaming. Furthermore, we explore both the immediate and time-related effects of AI across various consumption contexts.

2.4. Mindsponge theory

The mindsponge theory elucidates the information processing mechanisms within the human mind (Vuong, 2023). According to this theory, the personal processing of information is dynamic and balanced, where inappropriate values are eliminated while new ones are assimilated through a multi-filtering system. The absorption of information by an individual depends on two factors: availability and accessibility. Availability refers to the quality of information and its objective presence in the environment, while accessibility pertains to the cognitive capacity of the brain to assimilate or accept the available information.

The mindsponge theory offers an analytical framework that facilitates the construction of a psychological process by leveraging existing evidence (Vuong, 2023). Livestreaming e-commerce entails multifarious interactions between streamers and viewers (Xu & Ruan 2023). The utilization of AI in livestreaming enhances interaction with streamers, thereby increasing the acquisition of information by individuals. Simultaneously, the growing popularity of AI fosters a deeper understanding of AI-related information and facilitates access to high-quality AI resources over time, ultimately improving information availability. In accordance with the mindsponge theory, enhanced availability increases the likelihood of absorbing information and incorporating new values into one's mindset.

In terms of accessibility, the perceptual range of an individual exerts a significant influence on it. The perceptual range relates to what a person hears, sees, or senses (Nguyen, 2022), and these aspects can change over time. As the interaction with AI continues to deepen, changes in the individual's perceptual range improve the accessibility of information. People's new values are formed by assessing the cost and benefit of accepting or rejecting AI through the filtering system that enters the mindset.

According to the mindsponge theory, changes of information availability and accessibility over time in AI livestreaming will inevitably influence an individual's cognitive processes. Therefore, we developed models incorporating the time factor to quantify the effects of AI applications in livestreaming e-commerce, which have been lacking in previous research.

3. Data and methodology

3.1. Data sources and sample

To examine the effectiveness and explore the evolving role of AI streamers in livestreaming, we first developed a data crawling software using *Python* to collect livestreaming data from Taobao, the most influential livestream shopping platform in China, which provided a rich dataset for this study (In 2021, the total number of viewers on the Taobao livestreaming platform exceeded 50 billion.). After excluding samples featuring only human or AI livestreaming, our dataset included details from 21,190 livestreaming shows across 123 live rooms, covering the period from July 21 to December 9, 2021. Since products featured in livestreaming shows varied between live rooms, our dataset encompassed 924,036 products across various industries, such as beauty, office equipment, clothing, snacks, and more, involving both AI and human streamers.

For each livestreaming show, we collected data on the start and end times, the number of viewers, the frequency of viewer entries, the quantity of items listed, the output of promotional content, the average transaction value per customer, the volume of *danmaku* (viewer comments), the increase in followers, and the total sales. We determined the nature of the streamer's identity in each show—whether AI or human—by analysing the replay videos obtained from Taobao. Our main analysis was conducted at the level of the live room. The collected datasets are structured with a nested arrangement, where live rooms are the cross-sectional units and each livestreaming session within a live room is treated as a basic observation unit.

3.2. Variables

3.2.1. Dependent variables

The description of variables is shown in Table 1. In the dataset, two types of variable were used to measure the engagement: monetary engagement activities and non-monetary engagement activities per livestreaming session in each live room. The transaction volume (*Sales*), actual transaction volume (*Actsales*), and *pit output* were used as measures of monetary engagement activities. *Sales* represents the monetary value of sales per livestreaming show. *Actsales* refers to net sales, which are calculated by deducting product returns, and the *pit output* is an e-commerce term, which refers to the sales of a live room in the position under the category of an e-commerce platform. In addition to purchasing behaviours, viewers can tap the 'like' button to engage with the livestreamer (*NumLikes*), send *danmaku* (*NumDanmaku*) to interact with

Table 1
Variable description and measures.

Variable	Description	Measurement
streamer ID	Livestreamer ID.	String
Sales	The sales for every live room (refunds are not considered).	Integer
Actsales	The actual sales value per live room (refunds are considered).	Integer
PitOutput	The position under the category of e-commerce platform is a "pit." Pit output refers to the sales in a pit.	Integer
SaleVolume	The transaction quantity of merchandise in the livestreaming room.	Integer
AveValue	The average transaction value per customer.	Integer
NumItem	The number of items listed per live room	Integer
NumViewers	The number of viewers per live room.	Integer
NumVisits	The total number of visits to the live room.	Integer
Length	The duration of the live broadcast.	Integer
PeakTime	The duration of livestreaming during the peak period of users surfing the Internet.	Integer
NumLikes	The number of 'likes' received by the livestreamer in every live room.	Integer
NumDanmaku	The number of viewers chatting in the live comments by the viewers in every live room.	Integer
NumFollower	The number of increased followers in every live room.	Integer
AI	AI livestreamer.	Binary [1:AI; 0: Human]
Weekend	The livestreaming was launched on weekends.	Binary [1: Yes; 0: No]
Carnival	The livestreaming was launched during an online shopping carnival.	Binary [1: Yes; 0: No]
Attribution	The consumption type.	Binary [1: Utilitarian; −1: Hedonic]

other viewers and the livestreamer, and also follow the livestreamer (*NumFollower*) to show their affection or admiration. We also used these interactions to measure non-monetary engagement activities.

3.2.2. Independent variables

The identity of livestreamers was designated by using a binary indicator of AI and Human (AI = 1, Human = 0).

3.2.3. Control variables

Control variables, such as the number of viewers (*NumViewers*), the total number of times viewers entered the live room (*NumVisits*), the quantity of items per livestreaming session (*NumItem*), and the average transaction value per customer (*AveValue)* were considered. We also measured the total duration of each livestream (*Length*) and the specific duration during peak internet usage (*PeakTime*). Besides online shopping events, such as Alibaba's annual global online shopping carnival, Black Friday, weekends also have impacts on consumers' online behaviour. We included date controls to indicate whether the livestreaming occurred during a shopping carnival (*Carnival*) or on a weekend (*Weekend*).

3.3. Methodology

In line with prior studies that employed nested sampling or repeated measures, we applied the Linear Mixed Model (LMM) to estimate the factors contributing to viewer engagement (i.e., sales, pit output, liking, *danmaku* comments, or following). While a general linear regression would yield a more parsimonious model, the results can be biased if the assumptions of normally distributed data, homogeneity of variance, and independence of errors are violated (Bono et al., 2021). We therefore tested the results using the nested sampling method of the *nlme* R package (Pinheiro et al., 2022).

According to Raisch and Krakowski (2021), dual AI applications—automation and augmentation—are interdependent over time. To verify and explore the time-varying role of AI streamers in livestreaming, we used the TVEM model for estimation. Traditional methods, such as the dynamic linear model and Kalman filtering, often make strong assumptions about the underlying states and assume discrete time (Leeflang et al., 2009; Tan et al., 2012). However, in the absence of a pre-defined shape, TVEM models the coefficients as continuous smooth functions of time, thereby offering greater flexibility for exploring dynamic effects over time (Saboo et al., 2016). Specifically, different from the assumed constant parameter estimates (presented in Eq. (1)), TVEM allows for smooth variations of coefficients as a function of other variables, as depicted in Eq. (2). Therefore, in this study, we examined the evolving role of AI in livestreaming over time by employing the TVEM approach. Also, we used the SAS macro *%TVEM*,[9] developed by Li et al. (2017), to estimate the parameters.

$$y_{ij} = \beta_0 + \beta_1 x_{ij1} + \beta_2 x_{ij2} + \tau\cdots + \beta_p x_{ijp} + \varepsilon_{ij} \quad (1)$$

$$y_{ij} = \beta_0(t_{ij}) + \beta_1(t_{ij}) x_{ij1} + \beta_2(t_{ij}) x_{ij2} + \tau\cdots + \beta_p(t_{ij}) x_{ijp} + \varepsilon_{ij} \quad (2)$$

3.4. Model development

Fig. 2 illustrates how livestreamers—both AI and human—combined with consumption types (utilitarian and hedonic), affect viewer engagement. In Model 1, we formulated the viewer engagement model with livestreamers, control variables, and random effects for live studios. Model 2 extends Model 1 by incorporating the consumption type. Model 3 further extends Model 1 and Model 2 by incorporating the consideration of the time scale.

3.4.1. Model 1: Incorporating the type of livestreamer into the viewer engagement models

We investigated the impact of various factors on viewer engagement in every livestreaming show, including the livestreamer presenting it, the number of viewers, the frequency of viewers entering the live studio, the average transaction value per customer, the number of items, the transaction quantity, the duration of display, the length of time of the live broadcast during the peak period of internet usage, the weekend effect, the carnival effect, and more. The model specification is as follows:

$$\begin{aligned} Ln(P_{1ij}) = {} & \beta_1 Ln(NumViewers_{ij}) + \beta_2 Ln(NumVisit_{ij}) + \beta_3 Ln(AveValue_{ij}) \\ & + \beta_4 Ln(NumItem_{ij}) + \beta_5 Ln(SaleVolume_{ij}) + \beta_6 Ln\left(Length_{ij}\right) \\ & + \beta_7 Ln(PeakTime_{ij}) + \beta_8 AI + \beta_9 Weekend_{ij} + \beta_{10} Carnival_{ij} \\ & + \beta_{11} Ln(NumLikes_{ij}) \\ & + \beta_{12} Ln(NumDanmaku_{ij}) + \beta_{13} Ln(NumFollower_{ij}) + \mu_0 + \mu_{1i} \end{aligned} \quad (3)$$

$$\begin{aligned} Ln(P_{2ij}) = {} & \beta_0 + \beta_1 Ln(NumViewers_{ij}) + \beta_2 Ln(NumVisit_{ij}) \\ & + \beta_3 Ln(AveValue_{ji}) + \beta_4 Ln(NumItem_{ij}) + \beta_5 Ln\left(SaleVolume_{ij_{ij}}\right) \\ & + \beta_6 Ln\left(Length_{ij}\right) \\ & + \beta_7 Ln(PeakTime_{ij}) + \beta_8 AI + \beta_9 Weekend_{ij} + \beta_{10} Carnival_{ij} + \mu_0 + \mu_{2i} \end{aligned} \quad (4)$$

where P_{1ij} denotes the viewers' monetary engagement in a live studio i during the show j, such as sales, actual sales, and pit output. P_{2ij} denotes the viewers' non-monetary engagement in a live studio i in the show j, such as the number of 'likes', the volume of *danmaku*, and the number of

[9] The SAS macro %TVEM is available on https://methodology.psu.edu.

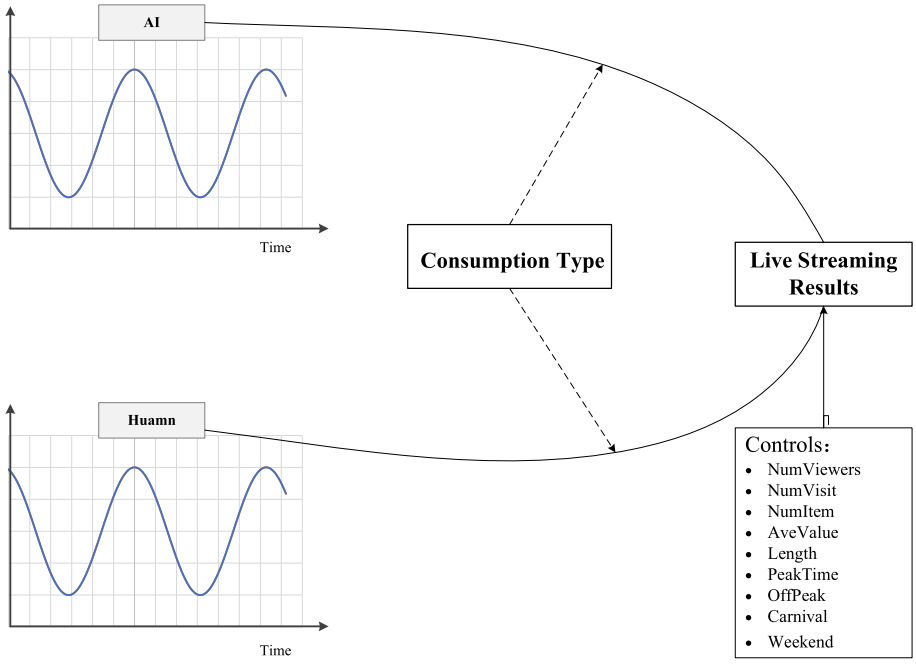


Fig. 2. Modelling framework Notes: Arcs represent dynamic curvilinear effects with time, straight arrows denote linear effects, and dashed arrows are moderating effects.

new followers. The parameters (μ_0, μ_i) represent the studio's fixed intercept and a random intercept, respectively. The random effects parsimoniously reflect the variability around the intercept μ_0 due to the heterogeneous impact of different studios.

3.4.2. Model 2: Incorporating consumption type

The proposed livestreaming analytics framework provides the consumption type for live studio *i* in show *j*, which we denote by $Attribution_{ij}$. Incorporating this into Model 2, we extended the right-hand side (RHS) of the equations in Model 1 as follows:

$$Ln(P_{1ij}) = \alpha_{11} Attribution_{ij} + \alpha_{12} Attribution_{ij} * AI + RHS(Equation3) \quad (5)$$

$$Ln(P_{2ij}) = \alpha_{21} Attribution_{ij} + \alpha_{22} Attribution_{ij} * AI + RHS(Equation4) \quad (6)$$

where α is the effect of consumption type on viewer engagement.

3.4.3. Model 3: Incorporating time Scales

Human expertise is the starting point of AI learning. Managers provide a machine with well-structured routine tasks, specifying the input and the corresponding outputs (Gillespie, 2022). From a process perspective, machine learning is not a static operation but an evolutionary process in which machines continually learn from data and continuously enhance their algorithms to improve their effectiveness in complex situations. Therefore, AI capabilities are dynamic and adapt to meet the needs of a changing work environment. For example, in Daugherty and Wilson's (2018) study, the decision to place Amelia (an AI agent) in the front office built on its successful first-phase deployment in an internal IT Service Desk supporting Skandinaviska Enskilda Banken's 15,000-strong workforce. This implies that the evolving nature of AI capabilities will continuously enhance AI effectiveness and efficiency in practice. To investigate such dynamic effects, we modified Model 2 to explore the continuously evolving nature of AI. Then, Model 3 is given by:

$$Ln(P_{1ij}) = (t_{ij}) RHS(Equation3) \quad (7)$$

$$Ln(P_{1ij}) = (t_{ij}) RHS(Equation4) \quad (8)$$

$$Ln(P_{1ij}) = (t_{ij}) RHS(Equation5) \quad (9)$$

$$Ln(P_{2ij}) = (t_{ij}) RHS(Equation6) \quad (10)$$

4. Analysis and findings

4.1. Descriptive statistics

The livestreaming shows in our sample had an average sale of above 1,000,000 *yuan (RMB)*, making them comparable to the global live studios on the Taobao platform. This suggests that our livestreaming selection does not introduce selection bias into our sample. Please refer to Table 2 for the descriptive statistics of the main variables.

Upon analysing the descriptive statistics of the main variables (Table 3), we observed differences between AI and human streamers in sales and interaction. These findings align with prior research, indicating that AI is suitable for well-structured tasks where the rules can be explicitly formulated and codified. The duration of each live studio for AI was 25.5 % longer than with human streamers. Given AI's weakness in interaction, there were fewer 'likes' (M_{likes} = 983.910) and *danmaku* ($M_{danmaku}$ = 57.720) for AI streamers compared with human beings (M_{likes} = 30,909.940, $M_{danmaku}$ = 1,196.560).

We also analysed the pairwise correlations among the main variables (see Table 4) to assess whether multicollinearity might be a potential source of endogeneity. The correlations did not exceed the conventional cut-off of 0.70 between variables in the same econometric model (Dormann et al., 2013). Although the correlation between sales and actual sales (0.996) was a notable exception, it is important to note that these variables were not included in the same econometric model.

4.2. Results

4.2.1. Engagement impact of livestreamer type

Krakowski et al. (2023) demonstrated that AI could substitute for human capability in chess tournament games, which differ from the business context. For example, players may opt to remain in a traditional niche that is no longer competitive (i.e., conventional chess)—a choice

Table 2
Descriptive statistics of main continuous variables for all samples.

	Obs.	Mean	SD	Min	Max
NumViewers	21,190	26,305.100	193,725.393	0	7,708,590
NumVisits	21,190	2,920.150	9,259.862	0	344,098
NumItem	21,190	41.590	43.617	1	500
Length	21,190	34,731.640	21,677.596	3	130,096
PeakTime	21,190	12,770.920	20,297.118	0	86,400
PitOutput	21,190	15,293.656	270,855.839	0	27,816,590
AveValue	21,190	524.656	1,121.099	0	28,888
NumLikes	21,190	19,511.500	524,242.159	0	54,333,749
NumDanmaku	21,159	763.900	2,611.038	0	83,872
NumFollower	21,149	181.130	979.170	0	39,079
SaleVolume	21,190	1,876.420	17,636.990	0	1,341,687
Sales	16,061	1,081,699.968	10,450,385.395	0	725,178,816
Actsales	10,833	1,351,739.260	12,405,051.382	0	724,886,208

Table 3
Summary of main livestreaming variables by livestreamer type.

	AI			Human		
	Obs.	Mean	SD	Obs.	Mean	SD
NumViewers	8071	3,823.240	14,123.139	13,119	4,0136.270	244,938.776
NumVisits	8071	186.340	572.339	13,119	4,602.020	11,439.928
NumItem	8071	28.680	33.307	13,119	49.530	47.167
Length	8071	39,731.180	23,652.355	13,119	31,655.840	19,749.565
PeakTime	8071	2,753.840	9,424.399	13,119	18,933.570	22,607.213
PitOutput	8071	8,525.790	123,637.025	13,119	19,457.346	330,228.605
AveValue	8071	450.860	994.830	13,119	570.057	1,189.921
NumLikes	8071	983.910	2,457.866	13,119	30,909.940	666,015.918
NumDanmaku	8071	57.720	147.746	13,119	1,196.560	3,241.083
NumFollower	8056	22.680	76.114	13,093	278.620	1,232.974
SaleVolume	8071	753.330	9,072.315	13,119	2,567.360	21,226.428
Sales	6205	420,033.406	5,760,117.678	9856	1,498,262.572	12,515,449.403
Actsales	4622	510,630.950	6,652,275.909	6211	1,977,661.430	15,315,733.775

Table 4
Matrix of pairwise correlations.

No	variables	1	2	3	4	5		7	8	9	10	11	12	13
1	NumViewers	1												
2	NumVisits	0.198	1											
3	NumItem	065	0.239	1										
4	Length	0.143	0.226	0.239	1									
5	PeakTime	0.185	0.370	0.226	−0.054	1								
6	PitOutput	0.092	0.101	0.370	0.093	0.478	1							
7	AveValue	−0.004	−0.023	0.101	−0.016	0.056	0.065	1						
8	NumLikes	0.058	0.055	−0.023	−0.030	−0.046	−0.034	0.057	1					
9	NumDanmaku	0.572	0.555	0.055	−0.003	0.001	0.023	0.030	−0.003	1				
10	NumFollower	0.758	0.346	0.555	0.133	0.246	0.370	0.207	0.001	0.065	1			
11	SaleVolume	0.176	0.214	0.042	0.084	0.113	0.511	−0.027	0.041	0.346	0.269	1		
12	Sales	0.191	0.217	0.346	0.073	0.177	0.242	0.136	−0.016	0.050	0.726	0.267	1	
13	Actsales	0.182	0.239	0.217	0.089	0.079	0.109	0.633	0.050	0.036	0.373	0.254	0.996	1

that is difficult to make in a market context. However, the relationship between AI and humans in market settings is less clear. Raisch and Krakowski (2021) proposed that the role of AI in management evolves over time, but the specifics of how it varies in particular tasks over time remain unclear. Our study addresses these questions. The estimates of −1.585, −1.482, and −0.540 in Table 5 (Model 1) indicate that the number of 'likes,' the number of *danmaku* comments, and the number of new followers decreased by 1.58 %, 1.48 %, and 0.54 %, respectively, for the AI streamer compared with the human streamer. This magnitude explains why retailers prefer to use human streamers during peak times, despite AI's capability to describe items in detail continuously. Shopping is not the only motivation for users watching livestreaming; emotional needs also play a significant role. The results in Table 5 also show no significant differences between AI and human beings in terms of pit output and final sales. This suggests that the observed substitution is a result of the introduction of AI.

A limitation of our analysis is that the absence of empirical differences between AI and humans in pit output and final sales does not conclusively demonstrate that no differences exist. Unlike the mixed linear regression testing we used, Bayesian analysis allowed us to glean evidence of absence, not just a mere lack of evidence (Keysers et al., 2020). Therefore, we conducted a Bayesian analysis to explore the potential non-differences further. The Bayes factor (BF10) results (Table 6) were 0.004 and 0.079, which are smaller than 0.01 and 0.1, respectively. These results suggest that AI and humans have no significant differences in pit output and final sales.

To explore the evolving role of AI in livestreaming, we employed the TVEM model in our study. Fig. 3 shows the estimation of all the time-varying covariates and their 95 % confidence intervals at various measurement times. Compared with human streamers, the impact of AI

Table 5
The Effects of Livestreamer Identity on Viewer Engagement.

	Sales		Actsales		PitOutput		NumLikes		NumDanmaku		NumFollower	
	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS
Intercept	3.444	2.798	4.244	3.804	1.316	1.106	−1.041	−0.237	−1.760	−1.900	−1.898	−2.164
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.033)	(0.000)	(0.000)	(0.000)	(0.000)
NumViewers	−0.103	−0.077	−0.143	−0.158	−0.001	0.018	0.287	0.309	−0.237	−0.141	−0.112	−0.067
	(0.000)	(0.000)	(0.000)	(0.000)	(0.665)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)
NumVisits	0.380	0.280	0.224	0.114	0.013	0.033	0.301	0.477	0.336	0.445	0.329	0.412
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
AveValue	0.966	0.999	0.974	0.983	0.738	0.796	0.011	0.010	−0.082	−0.042	−0.068	−0.035
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.048)	(0.088)	(0.000)	(0.000)	(0.000)	(0.000)
NumItem	−0.080	−0.064	−0.043	0.014	−1.004	−1.017	−0.151	−0.283	−0.020	−0.073	−0.039	−0.134
	(0.000)	(0.000)	(0.294)	(0.538)	(0.000)	(0.000)	(0.000)	(0.000)	(0.707)	(0.003)	(0.344)	(0.000)
SaleVolume	0.944	0.968	0.914	0.912	0.988	0.911	0.001	−0.024	0.018	−0.021	0.046	0.045
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.871)	(0.000)	(0.148)	(0.041)	(0.000)	(0.000)
Length	−0.327	−0.228	−0.433	−0.345	0.010	0.015	0.515	0.368	0.566	0.441	0.349	0.300
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
PeakTime	−0.052	−0.066	−0.080	−0.095	0.001	0.000	0.006	0.004	−0.001	−0.002	0.001	0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.016)	(0.382)	(0.002)	(0.039)	(0.775)	(0.603)	(0.696)	(0.754)
AI	−0.202	−0.383	−0.066	−0.225	0.023	−0.001	−1.585	−0.964	−1.482	−0.984	−0.540	−0.281
	(0.020)	(0.000)	(0.576)	(0.000)	(0.145)	(0.870)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Weekend	−0.035	0.014	−0.004	0.059	0.007	−0.010	0.006	−0.023	0.222	0.232	0.198	0.215
	(0.182)	(0.612)	(0.904)	(0.126)	(0.059)	(0.067)	(0.697)	(0.246)	(0.000)	(0.000)	(0.000)	(0.000)
Carnival	0.042	0.056	0.113	0.218	0.070	0.137	−0.272	−0.453	1.793	1.790	1.429	1.431
	(0.176)	(0.061)	(0.003)		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
NumLikes	−0.009	−0.033	−0.091	−0.149	0.002	0.002						
	(0.399)	(001)	(0.000)		(0.202)	(0.443)						
NumDanmaku	−0.039	−0.041	0.197	0.290	−0.004	−0.012						
	(0.006)	(0.002)	(0.000)		(0.057)							
NumFollower	0.042	0.039	0.224	0.238	0.001	0.011						
	(0.027)	(0.031)	(0.000)		(0.666)	(0.002)						

Notes: MLM means the estimated results of the mixed linear model; OLS represents the estimated results of ordinary linear regression.

Table 6
Bayesian Analysis (Relationship between AI and Humans).

	Full samples (Model 1)	Full samples (Model 2)	Utilitarian (sub-sample)	Hedonic (sub-sample)
Sales	—	0.029	0.350	—
Actsales	0.079	2.245	0.036	0.017
Pit Output	0.004	0.006	—	0.044
NumFollower	—	——	0.293	—

streamers on sales increased over time. Furthermore, due to the absence of human perception, emotion, and social skills (Braga & Logan, 2017), AI streamers did not perform well in viewer interactions, as evidenced by receiving fewer 'likes,' *danmaku*, and new followers, even over time.

4.2.2. Moderation effects of consumption type

Model 2 specifies the moderation effect of consumption type on the relationship between livestreamer type and viewer engagement. Table 7 shows that the estimated α is insignificant for most forms of engagement, in contrast to the findings reported by Longoni and Cian (2022). To further verify the role of AI in different consumption types, we divided the sample into utilitarian and hedonic contexts. The results are shown in Tables 8 and 9. In utilitarian consumption, AI and humans showed no significant differences in sales, although the pit output for AI streamers was higher than that for human beings. This suggests that the factors influencing consumer behaviour are relatively objective in the utilitarian consumption context. These factors can be easily identified and, consequently, improve the performance of AI.

We also conducted a Bayesian analysis to further explore the potential non-differences in the effect of livestreamer type on sales and final sales. The Bayes factor (BF10) results (Table 6) were 0.350 and 0.036, smaller than 1 and 0.1, respectively. These results suggest moderate evidence of the absence of no differences between AI and human streamers. However, human streamers performed better in *sales* (−0.292) in a hedonic context, although there was no difference in *pit output* and *Actsales* between the AI and the human live studio.[10] In addition, consistent with the research conclusions in Section 4.2.1, the number of 'likes,' *danmaku* comments, and new followers decreased in AI livestreaming in both consumption contexts.

To further understand why the results were inconsistent, it is important to investigate how the dynamic nature of interactions potentially influences these results. According to behavioural science research, incorporating the time factor is crucial for addressing process-oriented questions. This includes studying how a predictor is differentially associated with an outcome over time and making substantial theoretical contributions (Lanza et al., 2016).

Given that the prior literature indicates the dual application of AI (automation and augmentation) is interdependent over time, it is essential to consider this dynamic relationship. Augmentation may enable a transition to automation over time, and automation can provide people with more energy to enhance their performance on complex tasks. Therefore, we quantified the evolving effect of AI in livestreaming.

4.2.3. Time-varying effect of AI

This section presents the dynamic effectiveness of AI in different consumption contexts (see Figs. 4 and 5). As illustrated in Fig. 4, we found that the effect of AI streamers on pit output decreased over time and followed a nonlinear pattern in a utilitarian context. Compared to the coefficient of the AI streamer from the model without the time-varying effect ($\beta = 0.076$, $p < 0.001$), the TVEM suggests that the parameter estimate of the AI streamer decreased to $\beta = -0.075$ at the end of the time horizon. The effectiveness of AI streamers on sales was

[10] We also conducted a Bayesian analysis to explore the potential non-difference effect of livestreamer type on pit output and sales further. The Bayes factors (BF10) were 0.044 and 0.017, which are smaller than 0.1 and 0.3. These results suggest moderate evidence of the absence of no differences between AI and humans. These results suggest moderate evidence of no differences between AI and humans..

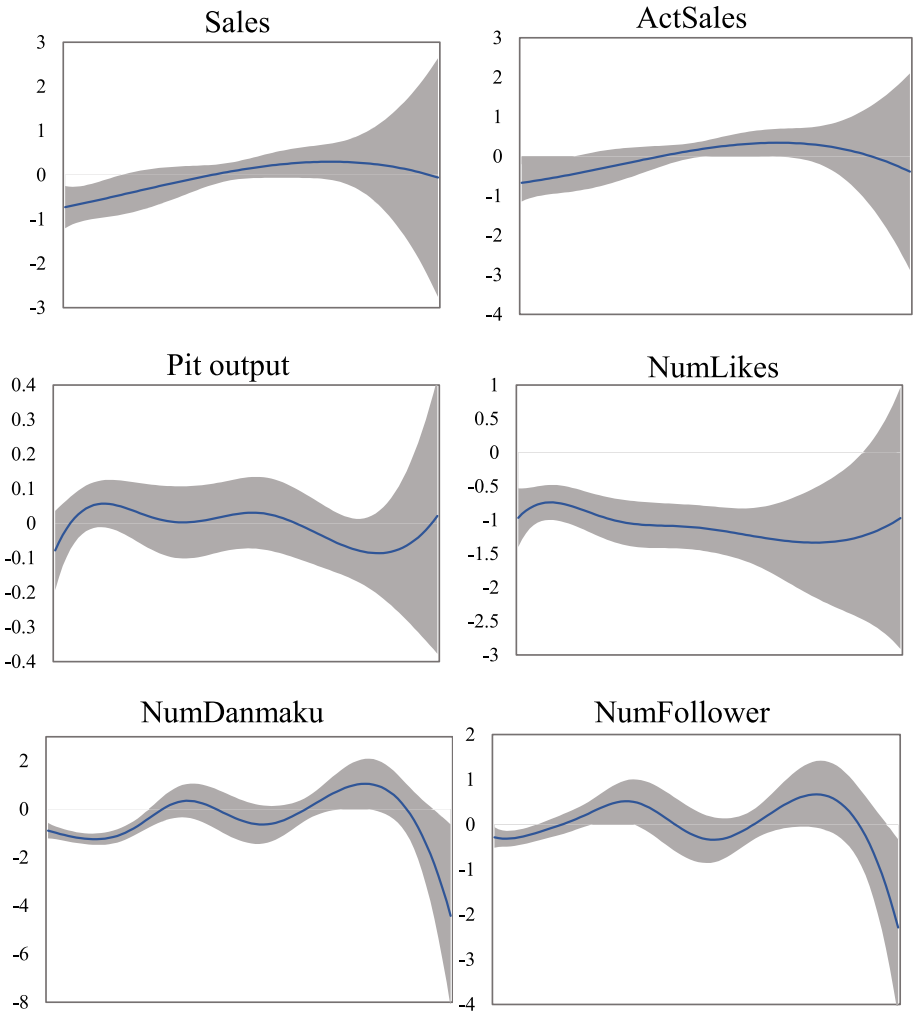


Fig. 3. The time-varying effect of AI on viewer engagement.

negative, remained flat, and did not change over time.

As Raisch and Krakowski (2021) discussed, organisations prefer to use AI for automation tasks due to its promise of short-term cost efficiencies. This strategy also inspires competitors to adopt AI to reduce costs. Consequently, the whole industry may be "entering a race to the zero-marginal reality" (Davenport & Kirby, 2016, p. 204), especially for rule-based tasks. Over time, early adopters of AI lose the low-cost advantage.

In addition to the monetary engagement activities, viewers can tap the 'like' button to express their attitude, appreciation, or other views about the livestreamer. It is well known that machines generally possess fewer perceptions, emotions, and social skills (Braga & Logan 2017). Therefore, it is understandable that the number of 'likes' obtained by AI livestreamers is low and does not change over time.

In a utilitarian consumption context, most questions and service requests pertain to the products (e.g., price, quality, size, material), which can be easily understood and follow relatively clear working rules. In livestreaming, AI streamers can effectively respond to these product-related inquiries, thus generating more interaction. As depicted in Fig. 4, the effect of AI on *danmaku* volume and the number of followers increases over time.

Fig. 5 depicts the sales impact of AI streamers in a hedonic context. First, we found that AI streamers positively affected sales. The difference with Fig. 4 is that our study found that the effectiveness of AI streamers on sales increased during the observation period rather than staying constant over time. These results may be related to the continuous iterative improvement of AI capabilities. In the iterative process, machines constantly learn new rules or create models and improve them over time (Rahwan et al., 2019). Consequently, AI streamers have an increasing effect on hedonic product sales.

Second, due to the absence of emotion and social skills, the effect of AI streamers on the number of 'likes' was negative and did not change over time. Finally, the curves in Fig. 4 show that AI streamers had a decreasing effect on *danmaku* volume and new followers over time. These results could be attributed to the highly personalized nature of questions and service requests in a hedonic context.

5. Discussion

5.1. Conclusions

Using empirical data from the Taobao platform, this study employed TVEM to investigate the time-varying impact of AI (versus human) streamers on viewer engagement in livestreaming across three distinct models. Our findings are outlined as follows: First, there was no discernible disparity in the performance of monetary activities between AI and human streamers, while human streamers exhibited a comparative advantage in non-monetary engagement activities. Second, AI was able to substitute for human streamers in monetary engagement activities in the context of utilitarian consumption, but not hedonic consumption. Third, the substitute effect of AI streamers was short-term in the utilitarian consumption context, which means that early adopters

Table 7
The Moderation Effects of Consumption Type with All Samples.

	Sales		Actsales		PitOutput		NumLikes		NumDanmaku		NumFollower	
	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS
Intercept	0.720	0.267	0.572	0.599	1.320	1.105	−1.847	−0.628	−2.509	−2.405	−2.166	−2.339
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
NumViewers	−0.017	0.003	−0.033	−0.022	−0.001	0.018	0.286	0.299	−0.237	−0.139	−0.112	−0.063
	(0.412)	(0.450)	(0.000)	(0.000)	(0.685)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)
NumVisits	0.033	−0.003	0.002	−0.001	0.013	0.032	0.301	0.481	0.336	0.446	0.329	0.411
	(0.037)	(0.580)	(0.792)	(0.921)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
AveValue	0.984	1.004	1.004	0.976	0.737	0.795	0.011	0.003	−0.082	−0.039	−0.068	−0.032
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.051)	(0.564)	(0.000)	(0.001)	(0.000)	(0.000)
NumItem	0.034	0.036	0.020	0.041	−1.003	−1.010	−0.148	−0.244	−0.017	−0.091	−0.040	−0.152
	(0.005)	(0.000)	(0.188)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.750)	(0.000)	0.340 ()	(0.000)
SaleVolume	0.999	1.006	1.000	0.990	0.988	0.913	0.001	−0.019	0.019	−0.024	0.046	0.043
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.858)	(0.000)	(0.143)	(0.021)	(0.000)	(0.000)
Length	−0.024	0.016	−0.004	−0.004	0.010	0.014	0.515	0.353	0.566	0.447	0.349	0.307
	(0.001)	(0.008)	(0.600)	(0.594)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
PeakTime	−0.003	−0.006	−0.007	−0.009	0.001	−0.001	0.006	0.002	−0.001	−0.002	0.001	0.002
	(0.003)	(0.000)	(0.000)	(0.000)	(0.017)	(0.328)	(0.002)	(0.238)	(0.770)	(0.668)	(0.691)	(0.587)
AI	−0.015	−0.050	−0.007	0.015	0.012	0.001	−0.787	−0.453	−0.742	−0.499	−0.271	−0.152
	(0.221)	(0.000)	(0.732)	(0.077)	(0.133)	(0.807)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Attribution	0.029	−0.001	−0.001	−0.006	−0.061	−0.021	−0.083	−0.069	−0.018	0.042	0.009	0.035
	(0.176)	(0.078)	(0.982)	(0.270)	(0.017)	(0.000)	(0.240)	(0.000)	(0.801)	(0.032)	(0.877)	(0.020)
AI*Attribution	−0.001	−0.014	−0.008	−0.016	0.002	0.000	0.042	0.132	−0.009	−0.020	−0.003	−0.051
	(0.849)	(0.002)	(0.482)	(0.003)	(0.703)	(0.880)	(0.349)	(0.000)	(0.853)	(0.271)	(0.921)	(0.000)
Weekend	−0.013	−0.010	−0.007	−0.004	0.007	−0.011	0.006	−0.021	0.222	0.232	0.198	0.215
	(0.123)	(0.278)	(0.499)	(0.724)	(0.060)	(0.060)	(0.699)	(0.292)	(0.000)	(0.000)	(0.000)	(0.000)
Carnival	0.035	0.035	0.035	0.049	0.070	0.137	−0.272	−0.446	1.793	1.788	1.429	1.428
	(0.001)	(0.001)	(0.003)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
NumLikes	0.007	−0.001	−0.006	−0.018	0.002	0.000						
	(0.055)	(0.710)	(0.169)	(0.000)	(0.208)	(0.830)						
NumDanmaku	−0.012	−0.027	0.026	0.049	−0.004	−0.012						
	(0.007)	(0.000)	(0.003)	(0.000)	(0.054)	(0.000)						
NumFollower	0.025	0.044	0.032	0.031	0.001	0.011						
	(0.000)	(0.000)	(0.000)	(0.000)	(647)	(0.002)						

Note: AI Utilitarian = 1; human, hedonic = −1.

Table 8
The Moderation Effects of Consumption Type in the Utilitarian Consumption Context.

	Sales		Actsales		PitOutput		NumLikes		NumDanmaku		NumFollower	
	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS
Intercept	4.253	3.195	4.615	4.094	1.153	0.984	−1.382	−0.647	−1.616	−1.596	−1.693	−2.018
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(004)	(0.000)	(0.000)	(0.000)
NumViewers	−0.066	−0.093	−0.192	−0.217	−0.001	0.025	0.336	0.351	−0.161	−0.125	−0.050	−0.043
	(0.229)	(0.000)	(0.000)	(0.000)	(0.803)	(0.000)	(0.000)	(0.000)	(0.031)	(0.000)	(0.423)	(0.029)
NumVisits	0.420	0.275	0.311	0.052	0.019	0.042	0.299	0.449	0.372	0.452	0.368	0.428
	(0.000)	(0.000)	(0.000)	(0.202)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
AveValue	0.947	0.964	0.984	0.947	0.752	0.810	0.007	−0.024	0.040	0.059	0.030	0.046
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.771)	(0.020)	(0.132)	(0.003)	(0.145)	(0.003)
NumItem	−0.118	−0.018	−0.074	0.123	−0.979	−0.978	−0.164	−0.141	0.084	−0.137	0.056	−0.115
	(0.047)	(0.531)	(0.304)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.248)	(0.000)	(325)	(0.000)
SaleVolume	0.952	0.985	0.950	0.945	0.993	0.906	−0.006	0.006	−0.058	−0.056	−0.019	0.002
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.462)	(0.420)	(0.001)	(0.000)	(0.164)	(0.857)
Length	−0.447	−0.234	−0.482	−0.340	−0.005	−0.001	0.508	0.312	0.397	0.392	0.200	0.239
	(0.000)	(0.000)	(0.000)	(0.000)	(0.135)	(0.820)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
PeakTime	−0.071	−0.086	−0.106	−0.121	0.001	0.002	0.004	0.011	0.008	−0.003	0.010	0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(030)	(0.037)	(0.150)	(0.000)	(0.244)	(617)	(0.047)	(0.899)
AI	−0.139	−0.637	−0.120	−0.518	0.076	0.057	−1.458	−0.527	−1.209	−1.016	−0.295	−0.305
	(0.414)	(0.000)	(0.578)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.072)	(0.000)
Weekend	−0.007	0.065	0.057	0.151	0.011	−0.011	−0.008	−0.071	0.208	0.230	0.173	0.207
	(0.870)	(0.184)	(0.329)	(0.017)	(0.009)	(0.177)	(0.744)	(0.022)	(0.000)	(0.000)	(0.000)	(0.000)
Carnival	0.026	0.200	0.054	0.300	0.049	0.107	−0.184	−0.431	1.748	1.751	1.334	1.377
	(0.610)	(0.000)	(0.374)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
NumLikes	−0.002	−0.054	−0.077	−0.176	0.003	−0.008						
	(0.910)	(0.001)	(0.002)	(0.000)	(0.093)	(0.006)						
NumDanmaku	−0.097	−0.109	0.103	0.234	−0.004	−0.002						
	(0.000)	(0.000)	(0.018)	(0.000)	(0.075)	(0.623)						
NumFollower	0.108	0.094	0.310	0.400	0.006	0.003						
	(0.001)	(0.002)	(0.000)	(0.000)	(0.072)	(0.567)						

Table 9
The Moderation Effects of Consumption Type in the Hedonic Consumption Context.

	Sales		Actsales		PitOutput		NumLikes		NumDanmaku		NumFollower	
	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS	MLM	OLS
Intercept	2.803	2.304	3.650	3.361	1.419	1.239	−0.868	0.312	−1.967	−2.152	−2.176	−2.229
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)	(0.037)	(0.001)	(0.000)	(0.000)	(0.000)
NumViewers	−0.065	−0.063	−0.105	−0.119	0.001	0.014	0.262	0.275	−0.290	−0.153	−0.151	−0.081
	(0.000)	(0.000)	(0.000)	(0.000)	(0.711)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
NumVisits	0.298	0.268	0.153	0.151	0.007	0.025	0.335	0.497	0.284	0.436	0.282	0.388
	(0.000)	(0.000)	(0.000)	(0.000)	(0.055)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
AveValue	0.983	1.015	1.001	1.002	0.725	0.784	0.018	0.013	−0.130	−0.089	−0.106	−0.078
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.149)	(0.059)	(0.000)	(0.000)	(0.000)	(0.000)
NumItem	−0.061	−0.020	0.011	0.025	−1.022	−1.049	−0.109	−0.317	−0.193	−0.099	−0.176	−0.234
	(0.097)	(0.357)	(0.816)	(0.433)	(0.000)	(0.000)	(0.003)	(0.000)	(0.014)	(0.006)	(0.003)	(0.000)
SaleVolume	0.941	0.961	0.889	0.901	0.984	0.927	0.003	−0.029	0.093	0.014	0.106	0.098
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.728)	(0.000)	(0.000)	(0.335)	(0.000)	(0.000)
Length	−0.261	−0.226	−0.395	−0.354	0.020	0.022	0.498	0.361	0.713	0.496	0.476	0.359
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
PeakTime	−0.038	−0.051	−0.058	−0.074	0.001	−0.002	0.005	−0.003	−0.007	−0.003	−0.005	0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.072)	(0.003)	(0.029)	(0.248)	(0.226)	(0.539)	(0.275)	(0.890)
AI	−0.292	−0.127	0.045	0.082	−0.020	−0.037	−1.602	−1.250	−1.790	−0.993	−0.811	−0.296
	(0.018)	(0.004)	(0.730)	(0.223)	(0.356)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Weekend	−0.055	−0.030	−0.053	−0.025	0.004	−0.009						
	(0.078)	(0.371)	(0.228)	(0.597)	(0.487)	(0.260)						
Carnival	0.057	−0.050	0.140	0.134	0.085	0.150						
	(0.124)	(0.160)	(0.003)	(0.003)	(0.000)	(0.000)						
NumLikes	−0.015	−0.017	−0.106	−0.129	0.001	0.004						
	(0.232)	(0.132)	(0.000)	(0.000)	(0.699)	(0.156)						
NumDanmaku	0.003	−0.001	0.270	0.319	−0.004	−0.012						
	(0.875)	(0.937)	(0.000)	(0.000)	(0.247)	(0.002)						
NumFollower	−0.008	0.010	0.155	0.132	−0.002	0.006						
	(725)	(0.634)	(0.000)	(0.000)	(0.643)	(0.251)						

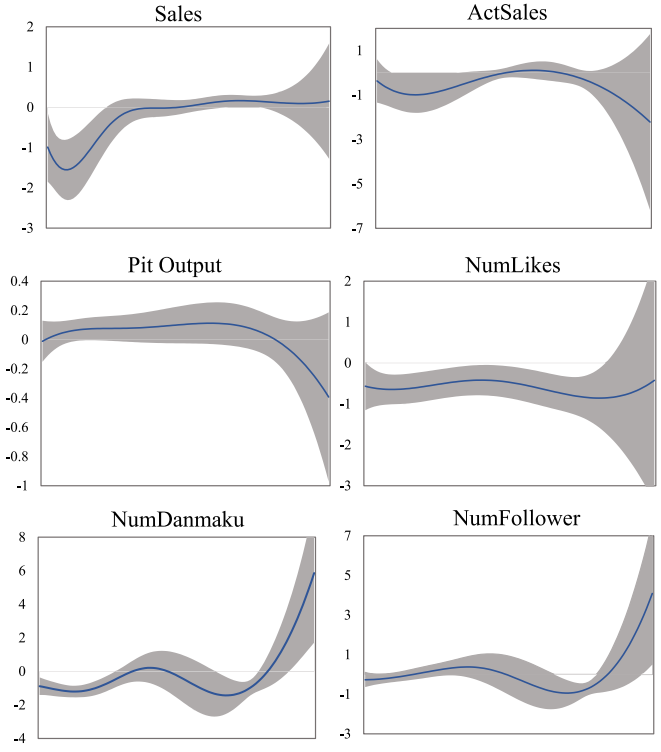


Fig. 4. The time-varying effect of AI on viewer engagement in the utilitarian context.

will lose the low-cost advantage as AI adoption becomes more widespread. However, AI streamers increasingly affected monetary engagement activities in the hedonic consumption context.

5.2. Theoretical contributions

This research has several theoretical contributions. The first set of contributions is related to the research on the relationship between AI and human beings. First, we expand this literature by investigating whether AI has the potential to surpass or outperform human capabilities in livestreaming e-commerce. Previous studies have primarily focused on identifying differences in consumer purchase intentions between AI and human streamers with a static perspective. By integrating the time factor proposed by Manis and Madhavaram (2023), this study investigates the dynamic relationship between AI and human streamers. Our work advances our understanding of whether AI can surpass humans in livestreaming. Thus, our research provides a new perspective on how the effects of AI and human streamers on viewer engagement vary over time. It thus responds to recent calls for more research on the relationships between AI and human (Krakowski et al., 2023; Raisch & Krakowski, 2021). Second, we demonstrate the circumstances under which AI streamers can outperform human beings in terms of utilitarian consumption through monetary engagement activities. Previous research in this area has largely focused on fixed-brand streamers, celebrity streamers, or virtual steamers. For example, it has examined how the salesperson's expressions, including happiness (Bharadwaj et al., 2022; Lin et al., 2021), and the characteristics of virtual streamers (Gao et al., 2023), impact sales and purchase intentions in livestreaming. Also, from the perspective of social relations, the social bonds, interactivity, and social status display initiated by human operators positively affect consumer engagement in livestreaming (Hou et al., 2020; Hu and Chaudhry, 2020). Limited research has identified the circumstances under which the performance of AI equals or exceeds human beings: if the utilitarian attribute is more important or salient in the product recommendation systems (Wien & Peluso 2021); if the product attribute is functional in online shopping assistance (Ruan & Mezei 2022); and if chatbot identity is not disclosed before the machine–customer conversation (Luo et al., 2019). We extend this literature by comparing how AI and human streamers trigger different forms of viewer engagement in livestreaming e-commerce. Third, we explore

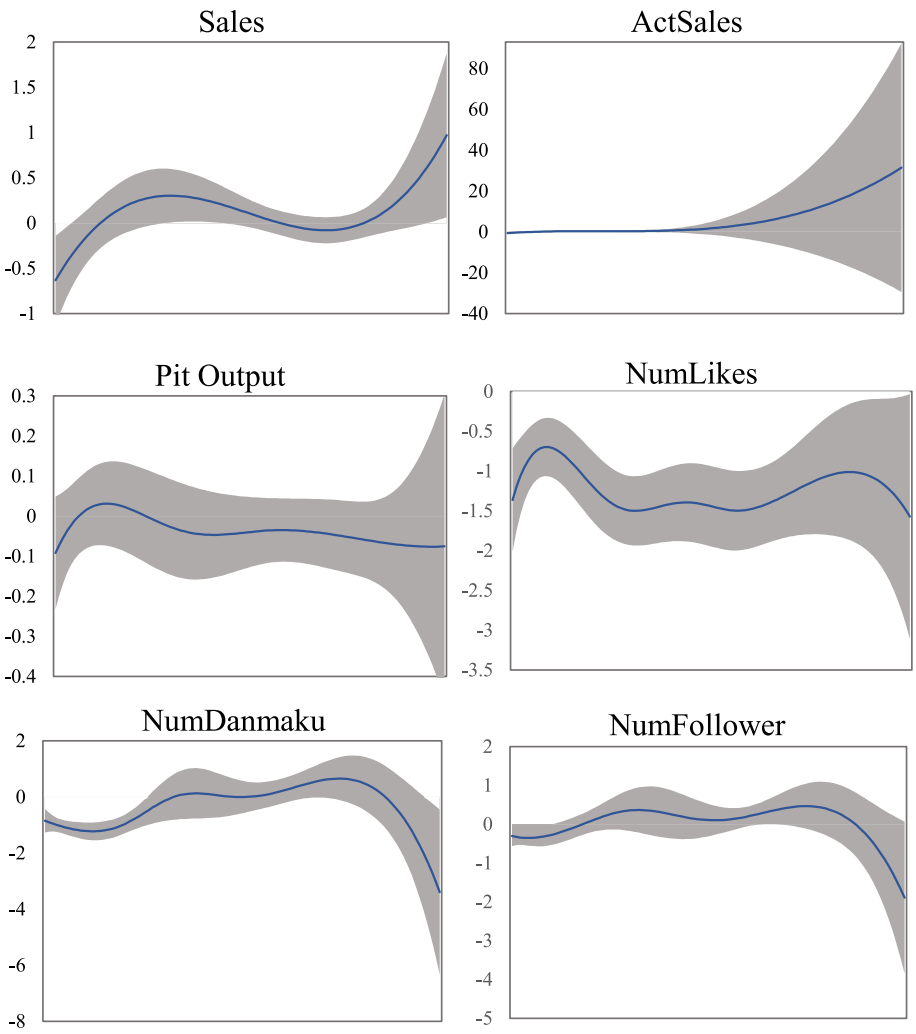


Fig. 5. The time-varying effect of AI on viewer engagement in the hedonic context.

when the performance of AI streamers is better than that of human beings and in which fields AI is more applicable. We demonstrate the substitution effect of AI streamers in the utilitarian context, which can also bring short-term performance improvement to enterprises. However, this effect can be temporary. AI is more suitable in the hedonic consumption context, even if it is not as effective as humans in the short term. This contribution is important because it is the first empirical study of the role of AI from an evolutionary process view. We expect that this contribution will inspire new research focused on understanding how the application potential of AI changes over time across consumption contexts. This is related to exploring the role of AI across time (Anthony et al., 2023; Vaid et al., 2023).

Secondly, we contribute to the literature on AI in marketing research. Literature in this area has focused on the characteristics and applicability of AI (Beeler et al., 2022; Blut et al., 2021), with the default presupposition that the ability of AI streamers does not change, or changes very slowly. We extend the literature by challenging this assumption and introducing the time-varying effect of AI from an evolutionary process view.

Finally, we contribute to research on AI streamers in both utilitarian and hedonic consumption contexts (Longoni & Cian, 2022; Wien & Peluso, 2021). The literature in this area has identified different recommendation effects of AI in utilitarian and hedonic consumption. We contribute to this literature by expanding the research scenario from recommendation systems to livestreaming e-commerce. Our examination of the future development potential of AI in different consumption contexts is approached from an evolutionary process perspective.

5.3. Managerial contributions

AI has revolutionized how companies engage with consumers and offers a powerful tool for optimizing livestreaming performance. However, for online retailers to enhance viewer engagement using this tool, they must understand when, in which fields, and on which engagement activities AI outperforms humans. Our findings reveal that the effectiveness of AI on viewer engagement activities varies over time. Therefore, online retailers should determine the appropriate consumption context before introducing AI streamers.

Our findings provide insights for online retailers on the most effective contexts for deploying AI streamers. Companies like Estée Lauder and McDonald's use both human and AI streamers. Our results suggest that AI streamers are more effective in utilitarian consumption contexts for enhancing monetary activities but may not be as effective in boosting non-monetary engagement activities.

While AI can substitute for human beings in utilitarian contexts for monetary activities, the advantage of this substitution effect may be limited. Our study suggests that online retailers could initially introduce AI streamers to enhance performance, with utilitarian attributes being more critical for the target segment. This aligns with other research (e.g., Raisch & Krakowski, 2021) which suggests that organizations pursue AI for short-term cost efficiencies. This strategy also encourages competitors to adopt AI, aiming to gain a competitive edge in cost efficiency.

Consequently, the initial cost advantage offered by AI may progressively diminish. For online retailers, maintaining a first-mover advantage with AI streamers becomes crucial for sustained success in the future.

While AI streamers may not have matched the performance of their human counterparts in the short term within the hedonic consumption context, this does not imply that AI lacks applicability. Our findings offer valuable insights for online retailers considering whether and when AI streamers would be more effective. This study highlights how AI streamers can increasingly enhance viewer engagement over time. Managers offering hedonic products should consider optimising or improving AI capabilities to achieve sustainable advantages.

5.4. Future research and limitations

While we have provided a detailed exploration of the evolving role of AI in livestreaming e-commerce, our research has limitations that present several opportunities for future study. First, although we examined the evolving role of AI streamers in two consumption contexts, variations in product categories might influence this role. For instance, introducing AI streamers for electronic products may offer greater short-term cost advantages compared to other categories (e.g., beauty, fashion). However, because the factors influencing purchasing behaviour are relatively objective and transparent, this advantage may dissipate more quickly than for other products. Future research should systematically explore the application of AI streamers across different product categories to investigate the boundary conditions of our findings.

Second, our study provides empirical evidence on the evolving role of AI without considering the types of AI. With technological advancements, AI now encompasses various forms of intelligence (e.g., mechanical AI, thinking AI, and feeling AI) (Huang & Rust, 2021b). Future studies could explore whether the role of different AI types evolves over time in livestreaming e-commerce, whether these changes vary under different AI conditions, and what capabilities or factors drive these changes.

Furthermore, our results are based on data from a single livestreaming platform. As livestreaming e-commerce continues to evolve, there are two primary modes: livestreaming embedded in e-commerce (e.g., Taobao) and e-commerce integrated into livestreaming (e.g., Douyin). Investigating whether our findings are replicable across different modes of livestreaming e-commerce would be a valuable avenue for future research.

CRediT authorship contribution statement

Haixia Yuan: Writing – original draft, Investigation, Data curation, Conceptualization. **Kevin Lü:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wenting Fang:** Data curation.

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Data availability

Data will be made available on request.

References

Anthony, C., Bechky, B. A., & Fayard, A.-L. (2023). "Collaborating" with AI: Taking a system view to explore the future of work. *Organization Science, 34*(5), 1672–1694. https://doi.org/10.1287/orsc.2022.1651

Balasubramanian, N., Ye, Y., & Xu, M. (2022). Substituting human decision-making with machine learning: Implications for organizational learning. *Academy of Management Review, 47*(3), 448–465. https://doi.org/10.5465/amr.2019.0470

Barta, S., Gurrea, R., & Flavián, C. (2023). Telepresence in live-stream shopping: An experimental study comparing Instagram and the metaverse. *Electronic Markets, 33*(1), 29. https://doi.org/10.1007/s12525-023-00643-6

Beeler, L., Zablah, A. R., & Rapp, A. (2022). Ability is in the eye of the beholder: How context and individual factors shape consumer perceptions of digital assistant ability. *Journal of Business Research, 148*, 33–46. https://doi.org/10.1016/j.jbusres.2022.04.045

Belanche, D., Casaló, L. V., Flavián, C., & Schepers, J. (2020a). Service robot implementation: A theoretical framework and research agenda. *Service Industries Journal, 40*(3–4), 203–225. https://doi.org/10.1080/02642069.2019.1672666

Belanche, D., Casaló, L. V., Flavián, C., & Schepers, J. (2020b). Robots or frontline employees? Exploring customers' attributions of responsibility and stability after service failure or success. *Journal of Service Management, 31*(2), 267–289. https://doi.org/10.1108/JOSM-05-2019-0156

Bharadwaj, N., Ballings, M., Naik, P. A., Moore, M., & Arat, M. M. (2022). A new livestream retail analytics framework to assess the sales impact of emotional displays. *Journal of Marketing, 86*(1), 27–47. https://doi.org/10.1177/00222429211013042

Bloomberg. (2020). Livestreams are the future of shopping in America. https://www.bnnbloomberg.ca/livestreams-are-the-future-of-shopping-in-america-1.1493619.

Blut, M., Wang, C., Wünderlich, N. V., & Brock, C. (2021). Understanding anthropomorphism in service provision: A meta-analysis of physical robots, chatbots, and other AI. *Journal of the Academy of Marketing Science, 49*(4), 632–658. https://doi.org/10.1007/s11747-020-00762-y

Bono, R., Alarcón, R., & Blanca, M. (2021). Report quality of generalized linear mixed models in psychology: A systematic review. *Frontiers in Psychology, 12*, Article 666182. https://doi.org/10.3389/fpsyg.2021.666182

Braga, A., & Logan, R. K. (2017). The emperor of strong AI has no clothes: Limits to artificial intelligence. *Information, 8*(4), 156. https://doi.org/10.3390/info8040156

Bughin, J., Seong, J., Manyika, J., Chui, M., & Joshi, R. (2018). Notes from the AI frontier: Modeling the impact of AI on the world economy. Retrieved from https://www.mckinsey.com/featured-insights/artificial-intelligence/notes-from-the-ai-frontier-modeling-the-impact-of-ai-on-the-world-economy. Accessed October 23, 2024.

Cai, J., & Wohn, D. (2019). *Live streaming commerce: Uses and gratifications approach to understanding consumers' motivations*. Hawaii International Conference on System Sciences, Hawaii.

Chiu, C.-M., Wang, E. T. G., Fang, Y.-H., & Huang, H.-Y. (2014). Understanding customers' repeat purchase intentions in B2C e-commerce: The roles of utilitarian value, hedonic value and perceived risk. *Information Systems Journal, 24*(1), 85–114. https://doi.org/10.1111/j.1365-2575.2012.00407.x

Choudhury, P., Starr, E., & Agarwal, R. (2020). Machine learning and human capital complementarities: Experimental evidence on bias mitigation. *Strategic Management Journal, 41*(8), 1381–1411. https://doi.org/10.1002/smj.3152

Daugherty, P. R., & Wilson, H. J. (2018). *Human + machine: Reimagining work in the age of AI*. Harvard Business Press.

Davenport, T. H., & Kirby, J. (2016). *Only humans need apply: Winners and losers in the age of smart machines*. HarperCollins.

Diwanji, V., Reed, A., Ferchaud, A., Seibert, J., Weinbrecht, V., & Sellers, N. (2020). Don't just watch, join in: Exploring information behavior and copresence on Twitch. *Computers in Human Behavior, 105*, 106221. https://doi.org/10.1016/j.chb.2019.106221.

Domingos, P. (2012). A few useful things to know about machine learning. *Communications of the ACM, 55*(10), 78–87. https://doi.org/10.1145/2347736.2347755

Dormann, C. F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., & Lautenbach, S. (2013). Collinearity: A review of methods to deal with it and a simulation study evaluating their performance. *Ecography, 36*(1), 27–46. https://doi.org/10.1111/j.1600-0587.2012.07348.x

Du, H., Niyato, D., Kang, J., Xiong, Z., Zhang, P., Cui, S., & Kim, D. I. (2024). The age of generative AI and AI-generated everything. *IEEE Network*. https://doi.org/10.48550/arXiv.2311.00947

Flavian, C., & Casaló, L. V. (2021). Artificial intelligence in services: Current trends, benefits and challenges. *Service Industries Journal, 41*(13–14), 853–859. https://doi.org/10.1080/02642069.2021.1989177

Gao, W., Jiang, N., & Guo, Q. (2023). How do virtual streamers affect purchase intention in the live streaming context? A presence perspective. *Journal of Retailing and Consumer Services, 73*, Article 103356. https://doi.org/10.1016/j.jretconser.2023.103356

Gao, J., Zhao, X., Zhai, M., Zhang, D., & Li, G. (2024). AI or human? The effect of streamer types on consumer purchase intention in live streaming. *International Journal of Human-Computer Interaction, 1–13*. https://doi.org/10.1080/10447318.2023.2299900

Garvey, A. M., Kim, T., & Duhachek, A. (2022). Bad news? Send an AI. Good news? Send a human. *Journal of Marketing, 87*(1), 10–25. https://doi.org/10.1177/00222429211066972

Gelbrich, K., Hagel, J., & Orsingher, C. (2021). Emotional support from a digital assistant in technology-mediated services: Effects on customer satisfaction and behavioral persistence. *International Journal of Research in Marketing, 38*(1), 176–193. https://doi.org/10.1016/j.ijresmar.2020.06.004

Gillespie, T. (2022). The relevance of algorithms. In T. Gillespie, P. J. Boczkowski, & K. A. Foot (Eds.), *Media technologies: Essays on communication, materiality, and society* (pp. 167–193). MIT Press.

Guan, Z., Hou, F., Li, B., Phang, C. W., & Chong, A.-Y.-L. (2022). What influences the purchase of virtual gifts in live streaming in China? A cultural context-sensitive model. *Information Systems Journal, 32*(3), 653–689. https://doi.org/10.1111/isj.12367

Guo, Y. Y., Zhang, K. X., & Wang, C. Y. (2022). Way to success: Understanding top streamer's popularity and influence from the perspective of source characteristics. *Journal of Retailing and Consumer Services, 64*, Article 102786. https://doi.org/10.1016/j.jretconser.2021.102786

Gursoy, D., Chi, O. H. X., Lu, L., & Nunkoo, R. (2019). Consumers acceptance of artificially intelligent (AI) device use in service delivery. *International Journal of Information Management, 49*, 157–169. https://doi.org/10.1016/j.ijinfomgt.2019.03.008

Hermann, E. (2022). Leveraging artificial intelligence in marketing for social good—An ethical perspective. *Journal of Business Ethics, 179*(1), 43–61. https://doi.org/10.1007/s10551-021-04843-y

Hilvert-Bruce, Z., Neill, J. T., Sjoblom, M., & Hamari, J. (2018). Social motivations of live-streaming viewer engagement on Twitch. *Computers in Human Behavior, 84*, 58–67. https://doi.org/10.1016/j.chb.2018.02.013

Hou, F., Guan, Z., Li, B., & Chong, A. Y. L. (2020). Factors influencing people's continuous watching intention and consumption intention in live streaming: Evidence from China. *Internet Research, 30*(1), 141–163. https://doi.org/10.1108/INTR-04-2018-0177

Hu, M., & Chaudhry, S. S. (2020). Enhancing consumer engagement in e-commerce live streaming via relational bonds. *Internet Research, 30*(3), 1019–1041. https://doi.org/10.1108/INTR-03-2019-0082

Huang, M.-H., & Rust, R. T. (2021a). Engaged to a robot? The role of AI in service. *Journal of Service Research, 24*(1), 30–41. https://doi.org/10.1177/1094670520902266

Huang, M.-H., & Rust, R. T. (2021b). A strategic framework for artificial intelligence in marketing. *Journal of the Academy of Marketing Science, 49*(1), 30–50. https://doi.org/10.1007/s11747-020-00749-9

Keysers, C., Gazzola, V., & Wagenmakers, E.-J. (2020). Author correction: Using Bayes factor hypothesis testing in neuroscience to establish evidence of absence. *Nature Neuroscience, 23*(11), 1453. https://doi.org/10.1038/s41593-020-00710-7

Khan, U., Dhar, R., & Wertenbroch, K. (2005). A behavioral decision theoretic perspective on hedonic and utilitarian choice. In S. Ratneshwar, & D. G. Mick (Eds.), *Inside Consumption: Consumer Motives, Goals, and Desires* (pp. 144–165). Routledge.

Kim, J., Kang, S., & Bae, J. (2022). The effects of customer consumption goals on artificial intelligence driven recommendation agents: Evidence from Stitch Fix. *International Journal of Advertising, 41*(6), 997–1016. https://doi.org/10.1080/02650487.2021.1963098

Krakowski, S., Luger, J., & Raisch, S. (2023). Artificial intelligence and the changing sources of competitive advantage. *Strategic Management Journal, 44*(6), 1425–1452. https://doi.org/10.1002/smj.3387

Lanza, S. T., Vasilenko, S. A., & Russell, M. A. (2016). Time-varying effect modeling to address new questions in behavioral research: Examples in marijuana use. *Psychology of Addictive Behaviors, 30*(8), 939–954. https://doi.org/10.1037/adb0000208

Leeflang, P., Bijmolt, T., van Doorn, J., Hanssens, D., van Heerde, H., Verhoef, P., & Wieringa, J. (2009). Creating lift versus building the base: Current trends in marketing dynamics. *International Journal of Research in Marketing, 26*, 13–20. https://doi.org/10.1016/j.ijresmar.2008.06.006

Li, J., Abbasi, A., Cheema, A., & Abraham, L. B. (2020). Path to purpose? how online customer journeys differ for hedonic versus utilitarian purchases. *Journal of Marketing, 84*(4), 127–146. https://doi.org/10.1177/0022242920911628

Li, R., Dziak, J. J., Tan, X., Huang, L., Wagner, A. T., & Yang, J. (2017). *TVEM (Time-Varying Effect Modeling) SAS Macro Users' Guide Version 3.1.1*. http://methodology.psu.edu.

Lim, J. S., Hwang, Y., Kim, S., & Biocca, F. A. (2015). How social media engagement leads to sports channel loyalty: Mediating roles of social presence and channel commitment. *Computers in Human Behavior, 46*, 158–167. https://doi.org/10.1016/j.chb.2015.01.013

Lin, Y., Yao, D., & Chen, X. (2021). Happiness begets money: Emotion and engagement in live streaming. *Journal of Marketing Research, 58*(3), 417–438. https://doi.org/10.1177/00222437211002477

Liu, F., Lim, E. T. K., Li, H., Tan, C.-W., & Cyr, D. (2020). Disentangling utilitarian and hedonic consumption behavior in online shopping: An expectation disconfirmation perspective. *Information & Management, 57*(3), Article 103199. https://doi.org/10.1016/j.im.2019.103199

Lo, P.-S., Dwivedi, Y. K., Wei-Han Tan, G., Ooi, K.-B., Cheng-Xi Aw, E., & Metri, B. (2022). Why do consumers buy impulsively during live streaming? A deep learning-based dual-stage SEM-ANN analysis. *Journal of Business Research, 147*, 325–337. https://doi.org/10.1016/j.jbusres.2022.04.013

Longoni, C., & Cian, L. (2022). Artificial intelligence in utilitarian vs. hedonic contexts: The "word-of-machine" effect. *Journal of Marketing, 86*(1), 91–108. https://doi.org/10.1177/0022242920957347

Lu, B. J., & Chen, Z. J. (2021). Live streaming commerce and consumers' purchase intention: An uncertainty reduction perspective. *Information & Management, 58*(7), Article 103509. https://doi.org/10.1016/j.im.2021.103509

Lu, S., Yao, D., Chen, X., & Grewal, R. (2021). Do larger audiences generate greater revenues under pay what you want? Evidence from a live streaming platform. *Marketing Science, 40*(5), 964–984. https://doi.org/10.1287/mksc.2021.1292

Luo, X., Qin, M. S., Fang, Z., & Qu, Z. (2021). Artificial intelligence coaches for sales agents: Caveats and solutions. *Journal of Marketing, 85*(2), 14–32. https://doi.org/10.1177/0022242920956676

Luo, X., Tong, S., Fang, Z., & Qu, Z. (2019). Frontiers: Machines vs. humans: The impact of artificial intelligence chatbot disclosure on customer purchases. *Marketing Science, 38*(6), 937–947. https://doi.org/10.1287/mksc.2019.1192

Lyytinen, K., Nickerson, J. V., & King, J. L. (2020). Metahuman systems = humans + machines that learn. *Journal of Information Technology, 36*(4), 427–445. https://doi.org/10.1177/0268396220915917

Ma, L., & Sun, B. (2020). Machine learning and AI in marketing–Connecting computing power to human insights. *International Journal of Research in Marketing, 37*(3), 481–504. https://doi.org/10.1016/j.ijresmar.2020.04.005

Manis, K. T., & Madhavaram, S. (2023). AI-enabled marketing capabilities and the hierarchy of capabilities: Conceptualization, proposition development, and research avenues. *Journal of Business Research, 157*, Article 113485. https://doi.org/10.1016/j.jbusres.2022.113485

Meng, L., Duan, S., Zhao, Y., Lü, K., & Chen, S. (2021). The impact of online celebrity in livestreaming E-commerce on purchase intention from the perspective of emotional contagion. *Journal of Retailing and Consumer Services, 63*, Article 102733. https://doi.org/10.1016/j.jretconser.2021.102733

Ming, J., Jianqiu, Z., Bilal, M., Akram, U., & Fan, M. (2021). How social presence influences impulse buying behavior in live streaming commerce? The role of S-O-R theory. *International Journal of Web Information Systems, 17*(4), 300–320. https://doi.org/10.1108/IJWIS-02-2021-0012

Nguyen, M. H. (2022). A new theory of serendipity: Nature, emergence and mechanism. In Q. H. Vuong (Ed.), *Conditions for improving serendipity encounter and attainment probability* (pp. 109–130). Sciendo.

Park, H. J., & Lin, L. M. (2020). The effects of match-ups on the consumer attitudes toward internet celebrities and their live streaming contents in the context of product endorsement. *Journal of Retailing & Consumer Services, 52*, Article 101934. https://doi.org/10.1016/j.jretconser.2019.101934

Pinheiro, J., Bates, D., DebRoy, S., Sarkar, D., authors, E., Heisterkamp, S., Team, R. C. (2022). *Package 'nlme'*. https://cran.r-project.org/web/packages/nlme/nlme.pdf.

Rahwan, I., Cebrian, M., Obradovich, N., Bongard, J., Bonnefon, J.-F., Breazeal, C., & Wellman, M. (2019). Machine behaviour. *Nature, 568*(7753), 477–486. https://doi.org/10.1038/s41586-019-1138-y

Raisch, S., & Krakowski, S. (2021). Artificial intelligence and management: The automation–augmentation paradox. *Academy of Management Review, 46*(1), 192–210. https://doi.org/10.5465/amr.2018.0072

Ruan, Y., & Mezei, J. (2022). When do AI chatbots lead to higher customer satisfaction than human frontline employees in online shopping assistance? Considering product attribute type. *Journal of Retailing and Consumer Services, 68*, Article 103059. https://doi.org/10.1016/j.jretconser.2022.103059

Puntoni, S., Reczek, R. W., Giesler, M., & Botti, S. (2021). Consumers and artificial intelligence: An experiential perspective. *Journal of Marketing, 85*(1), 131–151. https://doi.org/10.1177/0022242920953847

Saboo, A., Kumar, V., & Park, I. (2016). Using big data to model time-varying effects for market resource (re)allocation. *MIS Quarterly, 40*(4), 911–939. https://doi.org/10.25300/MISQ/2016/40.4.06

Smith, B. C. (2019). *The promise of artificial intelligence: Reckoning and judgment*. MIT Press.

Tan, X., Shiyko, M. P., Li, R., Li, Y., & Dierker, L. (2012). A time-varying effect model for intensive longitudinal data. *Psychological Methods, 17*(1), 61–77. https://doi.org/10.1037/a0025814

Vaid, S., Puntoni, S., & Khodr, A. (2023). Artificial intelligence and empirical consumer research: A topic modeling analysis. *Journal of Business Research, 166*, Article 114110. https://doi.org/10.1016/j.jbusres.2023.114110

Venkatesh, V., Morris, M., Davis, G., & Davis, F. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly, 27*, 425–478. https://doi.org/10.2307/30036540

Vuong, Q.-H., La, V.-P., Nguyen, M.-H., Jin, R., La, M.-K., & Le, T.-T. (2023a). AI's humanoid appearance can affect human perceptions of its emotional capability: Evidence from self-reported data in the U.S. *International Journal of Human-Computer Interaction, 40*(17), 4906–4917. https://doi.org/10.1080/10447318.2023.2227828

Vuong, Q.-H., La, V.-P., Nguyen, M.-H., Jin, R., La, M.-K., & Le, T.-T. (2023b). How AI's self-prolongation influences people's perceptions of its autonomous mind: The case of U.S. residents. *Behavioral Sciences, 13*, 470. https://doi.org/10.3390/bs13060470

Vuong, Q.-H. (2023). Mindsponge theory. *De Gruyter*. https://doi.org/10.2478/9788367405157

Wang, M., Shan, G., & Thatcher, J. (2022). Human versus AI? Investigating the heterogeneous effects of live streaming e-commerce. *Proceedings of AMCIS 2022, Minnesota*.

Wang, W., Huang, M., Zheng, S., Lin, L., & Wang, L. (2021). The impact of broadcasters on consumer's intention to follow livestream brand community. *Frontiers in Psychology, 12*, Article 810883. https://doi.org/10.3389/fpsyg.2021.810883

Wien, A. H., & Peluso, A. M. (2021). Influence of human versus AI recommenders: The roles of product type and cognitive processes. *Journal of Business Research, 137*, 13–27. https://doi.org/10.1016/j.jbusres.2021.08.016

Wongkitrungrueng, A., & Assarut, N. (2020). The role of live streaming in building consumer trust and engagement with social commerce sellers. *Journal of Business Research, 117*, 543–556. https://doi.org/10.1016/j.jbusres.2018.08.032

Wongkitrungrueng, A., Dehouche, N., & Assarut, N. (2020). Live streaming commerce from the sellers' perspective: Implications for online relationship marketing. *Journal of Marketing Management, 36*(5/6), 488–518. https://doi.org/10.1080/0267257X.2020.1748895

Xu, Y. J., & Ruan, Y. Y. (2023). AI and human broadcasters: Relative impact on consumer engagement in live streaming commerce. *Electronic Commerce Research and Applications, 62*, Article 101335. https://doi.org/10.1016/j.elerap.2023.101335

You, S., & Robert, L. P. (2023). Trusting and working with robots: A relational demography theory of preference for robotic over human co-workers. *MIS Quarterly (forthcoming)*. https://doi.org/10.7302/7223

Zhang, L. T., & Cassany, D. (2020). Making sense of danmu: Coherence in massive anonymous chats on bilibili.com. *Discourse Studies, 22*(4), 483–502. https://doi.org/10.1177/1461445620940051

Zhang, M., Liu, Y., Wang, Y., & Zhao, L. (2022). How to retain customers: Understanding the role of trust in live streaming commerce with a socio-technical perspective. *Computers in Human Behavior, 127*, Article 107052. https://doi.org/10.1016/j.chb.2021.107052

Zhao, K., Hu, Y., Hong, Y., & Christopher, W. J. (2021). Understanding characteristics of popular streamers on live streaming platforms: Evidence from Twitch.tv. *Journal of the Association for Information Systems, 22*(4), 1076–1098. https://doi.org/10.17705/1jais.00689

Zhang, Y., Wang, X., & Zhao, X. (2023). Supervising or assisting? The influence of virtual anchor driven by AI–human collaboration on customer engagement in live streaming e-commerce. *Electronic Commerce Research, 2023*. https://doi.org/10.1007/s10660-023-09783-5

Zhang, X., Shi, Y., Li, T. T., Guan, Y., & Cui, X. (2023). How do virtual AI streamers influence viewers' livestream shopping behavior? The effects of persuasive factors and the mediating role of arousal. *Information Systems Frontiers, 2023*. https://doi.org/10.1007/s10796-023-10425-2

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